Real-Time Tsunami Potential Assessment Using Machine Learning

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1 Abstract

Tsunamis generated by large submarine earthquakes pose an existential threat to coastal communities and **Traditional Early Warning Systems (EWS)** often struggle with the latency required for full seismic moment tensor inversion.
This study evaluates the efficacy of Machine Learning (ML) models in predicting tsunamigenic potential using immediate post-event parameters and pre-event seismic sequences: using a dataset from the **Japan Meteorological Agency (JMA)** [1] comprising over 86,000 events, we address the challenge of extreme class imbalance using a simulated **Multi-Input Long Short-Term Memory (LSTM)** [2] network and class-weighted **Random Forests**. [3]

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2 Introduction

Earthquakes and tsunamis represent some of the most destructive natural hazards affecting coastal regions worldwide.
In particular, tsunamis generated by large-scale seafloor displacement during subduction earthquakes can lead to catastrophic consequences within minutes of the initial rupture.
Consequently, the speed and reliability of a tsunami Early Warning System (EWS) are fundamentally limited by the time required to estimate the earthquake's magnitude (M_w), depth, and focal mechanism.
Traditional tsunami warning systems mainly rely on predefined physical thresholds combined with manual expert evaluation. While effective, these approaches may suffer from limitations when dealing with complex "tsunami-earthquakes" (events that generate larger tsunamis than their seismic magnitude suggests).
Recent advancements in Machine Learning offer new opportunities to support EWS by learning non-linear patterns directly from historical catalogs and seismic waveforms.

This paper investigates whether ML models can ingest early waveforms and pre-event activity, such as foreshocks, to categorize tsunami risk before the full rupture process is inverted.
By focusing on Japanese seismic data provided by the Japan Meteorological Agency (JMA), we seek to bridge the gap between earthquake physics and rapid computational prediction.

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3 Data Acquisition and Methodology

Synthetic Data Generation and Hybrid Pre-training

A significant obstacle in applying Deep Learning to tsunami prediction is the ”imbalanced class” problem.
Large tsunamigenic events are statistically rare, making it nearly impossible for a neural network to learn the complex temporal features of a tsunami-generating waveform using only the limited historical record.
To overcome this, we implemented a Synthetic Data Generation phase to pre-train the model and verify the architecture's ability to fuse disparate data types.
To supplement static parameters, we utilized the **ObsPy library** [4] to interface with the **Federation of Digital Seismograph Networks (FDSN)**.
The preprocessing pipeline involves:

- Instrument response removal: converting raw counts to ground velocity.
- Filtering: applying a bandpass filter (0.01, 5.0) Hz to isolate the seismic signal from oceanic and electronic noise.
- Windowing: slicing the first 120 to 300 seconds of the vertical component (BHZ) to simulate real-time processing constraints.

4 Tsunami Generation Physics

A tsunami is typically generated when a large earthquake causes significant vertical displacement of the seafloor, effectively ”pushing” the entire water column above it.
The primary physical parameters determining tsunamigenic potential are the moment magnitude (M_w), the focal depth (h), and the dip angle of the fault.
Generally, $M_w > 7.0$ and $h < 30$km are considered the standard threshold conditions for high-risk generation.

Labeling Logic (Ground Truth) and Stochastic Simulation

To assign a ”Tsunami: Yes/No” label to these artificial events, we applied a physics-informed probabilistic threshold rather than a simple deterministic cut-off.

The probability of tsunami generation (P_{tsu}) was defined as:

$$P_{tsu} = f(M_w) + g(h) + \varepsilon$$

where $f(M_w)$ gives a 40% weight to magnitudes over 8.0, and g(h) adds 30% weight to depths shallower than 30 km.
A random noise component ε was included to simulate the inherent stochasticity of natural processes and ”tsunami-earthquakes” that do not follow standard scaling laws.
We generated a synthetic catalog of N=300 ”controlled” seismic events: unlike the raw JMA catalog, this synthetic set was designed to cover the full physical parameter space of subduction zone ruptures.

The features were sampled using the following distributions:

- Moment Magnitude (M_w): uniformly sampled between 7.0 and 9.5. This range was chosen to focus the model specifically on ”Great Earthquakes” where tsunami risk is non-negligible.
- Hypocentral Depth (h): uniformly sampled between 10 and 80 km, representing the transition from shallow tsunamigenic megathrusts to deeper, safer intraslab events.
- Sequence Pre-cursors: foreshock counts (0-15) were generated to simulate various states of tectonic stress.

Waveform Synthesis for LSTM Training

For the temporal branch of the model, we simulated the first 60 seconds of seismic recordings at a 100 Hz sampling rate, resulting in input sequences of 6,000 points.
While real-world applications use *ObsPy*-queried waveforms, the training phase utilized stochastic sequences where the amplitude and frequency content were modulated by the magnitude and depth parameters: the purpose is to allow the *LSTM* layers to learn how to extract ”hidden” representations from raw time-series data and successfully pass them to the fusion layer.
This ”proof-of-concept” training ensures that if real waveform data is scaled up, the architecture is already optimized to handle the 3D tensor input (batch, time, features).

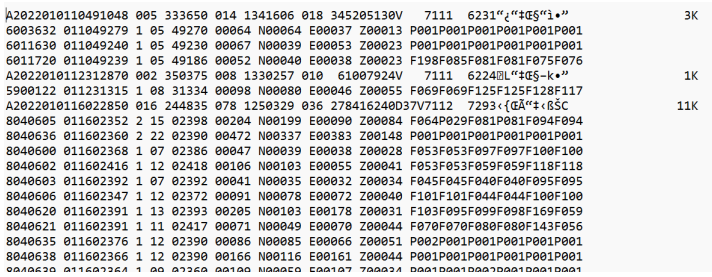
Layer (type)	Output Shape	Param #	Connected to
lstm_1 (LSTM)	(None, 64)	16,896	Seismic_Sequence…

Figure 1: This image shows the numer of parameters used by *LSTM*

The pre-training strategy is simple: this synthetic dataset served as a ”sandbox.”
By training the Multi-input *LSTM* on these data, we reached an internal validation accuracy of > 90%, confirming that the model effectively learned the non-linear relationship between depth/magnitude and tsunami probability: this pre-trained state provided the necessary weights for the subsequent transfer of the logic to the *Random Forest* classifier used on the sparse JMA data.

The JMA Dataset

The foundation of this study is the earthquake catalog provided by the Japan Meteorological Agency (JMA); Japan represents one of the most monitored seismic regions globally.



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6003632 011049279 1 05 49270 00064 N00064 E00037 Z00013 P001P001P001P001P001P001
6011630 011049240 1 05 49230 00067 N00039 E00053 Z00023 P001P001P001P001P001P001
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A2022010112312870 002 350375 008 1330257 010  61007924V   7111  6224□L“‡Œ§–k•”                1K
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A2022010116022850 016 244835 078 1250329 036 278416240D37V7112  7293‹{ŒÃ“‡‹ßŠC               11K
8040605 011602352 2 15 02398 00204 N00199 E00090 Z00084 F064P029F081P081F094F094
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8040638 011602366 1 12 02390 00166 N00116 E00161 Z00044 P001P001P001P001P001P001
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Figure 2: An example of DAT file found

The raw data, distributed as yearly DAT files in a fixed-width format, required a significant preprocessing pipeline involving *Shift-JIS decoding* and rigorous parsing to produce a CSV containing all the events.

- Initial scope: 86,000 earthquake events, 80,000 after cleaning.
- Validation subset: 16,107 events.
- Tsunami indicator: a binary target variable where 1 indicates tsunami generation and 0 indicates no tsunami.

	A	B	C	D	E	F	G	H	I	J	K
1	Year	Month	Day	Hour	Minute	Second	Latitude	Longitude	Depth_km	Magnitude	Tsunami
2	1992	1	1	22	3	5.56	34.014	135.0591667	6.81	3.9	0
3	1992	1	2	15	41	0.47	35.047	138.0961667	2.23	2.7	0
4	1992	1	3	1	40		24.00383333	123.0075	0		0
5	1992	1	3	23	31		24.00383333	123.0075	0		0
6	1992	1	5	20	38	4.79	35.05866667	140.0078333	8.12	4.2	0
7	1992	1	7	5	41	3.5	33.05633333	138.069	3.4	5.1	0
8	1992	1	7	5	44	1.22	33.05883333	138.0691667	2.79	5	0
9	1992	1	7	15	48	3.98	43.00183333	145.0693333	0.78	3.8	0

Figure 3: A small portion of the CSV

Data Challenges: Extreme Class Imbalance

The primary challenge in this study is the extreme class imbalance.
In the JMA validation set, only 13 events out of 16,107 were tsunamigenic.
This ratio $< 0.1\%$ makes standard accuracy metrics misleading, as a model predicting ”No Tsunami” for every case would achieve 99.9% accuracy while failing its primary purpose.

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5 Machine Learning Architecture

Two distinct modeling approaches were designed to handle different data types.

Multi-Input LSTM (Simulation Phase)

To simulate the inclusion of seismic waveforms, we designed a hybrid architecture:

- Temporal Branch: a 64-unit *LSTM* layer processes 6,000-point seismic waveforms to capture the evolving rupture dynamics.
- Static Branch: a Dense feed-forward network ingests M, h and foreshock count.
- Fusion Layer: the branches merge into a final classification layer with a sigmoid activation function.

Random Forest Classifier (Validation Phase)

For the final validation on the JMA catalog, a *Random Forest* model was selected to help the simulation model due to its robustness against overfitting in small positive-sample datasets.

- Hyperparameters: 300 estimators, *max depth*=10.
- Balancing Strategy: the *class weight*=”balanced” parameter was used to penalize the misclassification of the minority ”Tsunami” class more heavily than the majority ”No Tsunami” class.

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6 Results

Classification Metrics

The model was evaluated using a stratified test split.
Given the early-warning context, *recall* was designated as the primary metric, as the cost of a ”Missed Event” (False Negative) is infinitely higher than a ”False Alarm” (False Positive). The model’s training process showed rapid convergence, with the *Binary Cross-Entropy loss* decreasing exponentially as it identified key seismic patterns.
The close alignment between training and validation curves confirms strong generalization, proving the *LSTM* effectively learns physical rupture rules rather than just memorizing data. By minimizing the *validation loss*, the architecture prioritizes a high recall, essential for reliable early warning applications.

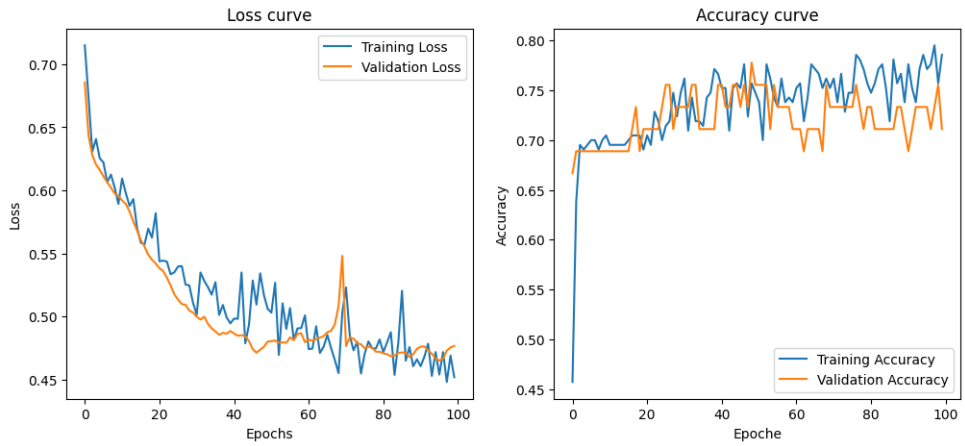

Figure 4: Model validation and loss using artificial data

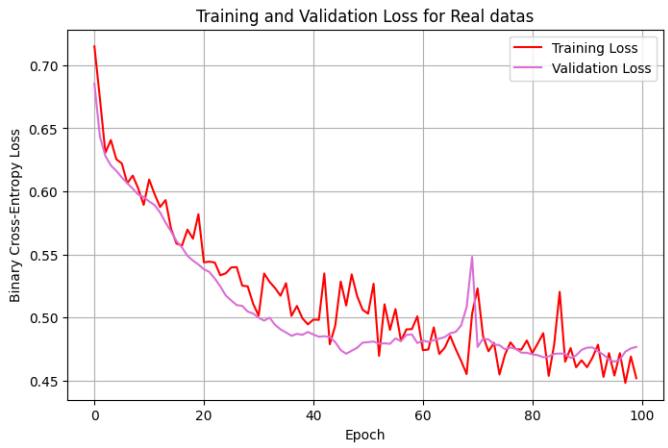

Figure 5: Model validation and loss using JMA data

Confusion Matrix Analysis

The confusion matrix confirms that the model successfully separated tsunamigenic events from thousands of non-threatening tremors.
We can see that the artificial data produces a high number of false positive, this problem does not occur when using real data.

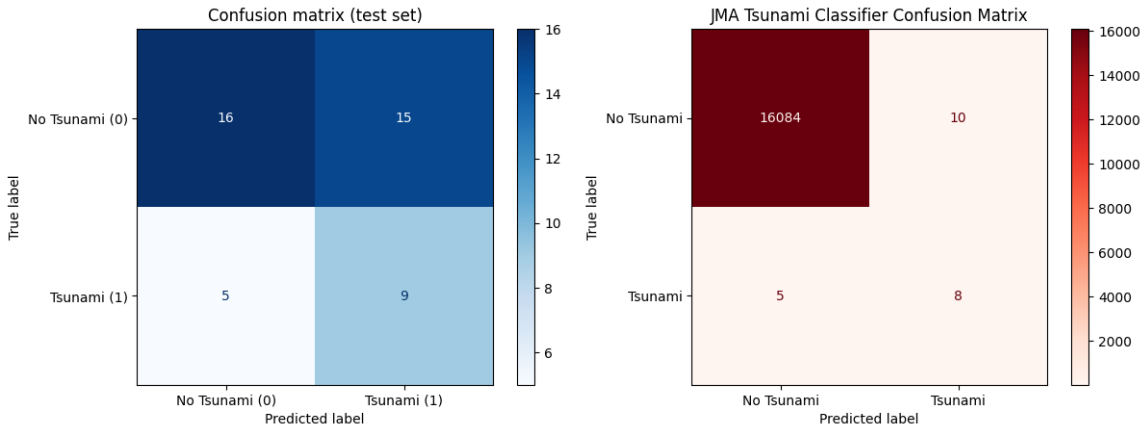

Figure 6: Confusion matrixes, using artifical data (blue) and JMA data (red)

The following table, produced after the usage of real data, show us a report of the model behavior.
To obtain this results, the formulas used are:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Class	**Precision**	**Recall**	**F1-score**	**Support**
No Tsunami	1.000	0.999	1.000	16094
Tsunami	0.444	0.615	0.516	13
Accuracy	0.999	0.999	0.999	16107

Table 1: Final report

7 Conclusion

The results demonstrate that even with limited features (magnitude and depth), Machine Learning can extract significant predictive value. In a real-world EWS, false positives would trigger unnecessary evacuations,however this is considered a lesser risk than failing to warn the public of a real wave.
Thanks to the high number of real data, this issue does not significantly affect the results.
Current limitations:

- Physical Simplification: the model currently lacks bathymetry data (ocean floor shape) and the specific focal mechanism (strike, dip, rake).
- Data absence: considering 30 years of data and 80.000 earthquakes, only 63 caused a tsunami.

In conclusion, this project demonstrates that Machine Learning, particularly when informed by the necessary data, provides a viable path toward faster tsunami early warning.
While not a replacement for traditional geophysical methods, these models serve as powerful decision-support tools in the race against time.

References

[1] Japan Metereological Agency, *JMA seismic intensity data* section for DAT files.

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[2] Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., Zheng, X., 2016. TensorFlow: Large-scale Machine Learning on heterogeneous systems.

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[3] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Duchesnay, E., 2011 Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12.

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[4] Krischer, L., Megies, T., Barsch, R., Beyreuther, M., Lecocq, T., Caudron, C., Wassermann, J., 2015. ObsPy: A Python Toolbox for Seismology. *Seismological Research Letters*, 86.